

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5112

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Annexure A

CONCEPT MAPPING

Code of Conduct:

I C Sangita(192373059) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Csangita

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability). Perform the following tasks:
 - a. Show how each attack leads to specific threats and risks
 - b. Map how security goals help mitigate them Provide one real-world example for each link
2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.
3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
 - a. Show differences and similarities
 - i. Map how each affects the Data structure and Security level
 - ii. Analyze which is more vulnerable to attacks and why
4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
 - a. Show relationships between these concepts
 - b. Analyze how each contributes to secure communication
 - c. Include one real-world application for each
5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
 - a. Show how each concept is mathematically connected
 - b. Map their role in encryption/decryption
 - c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

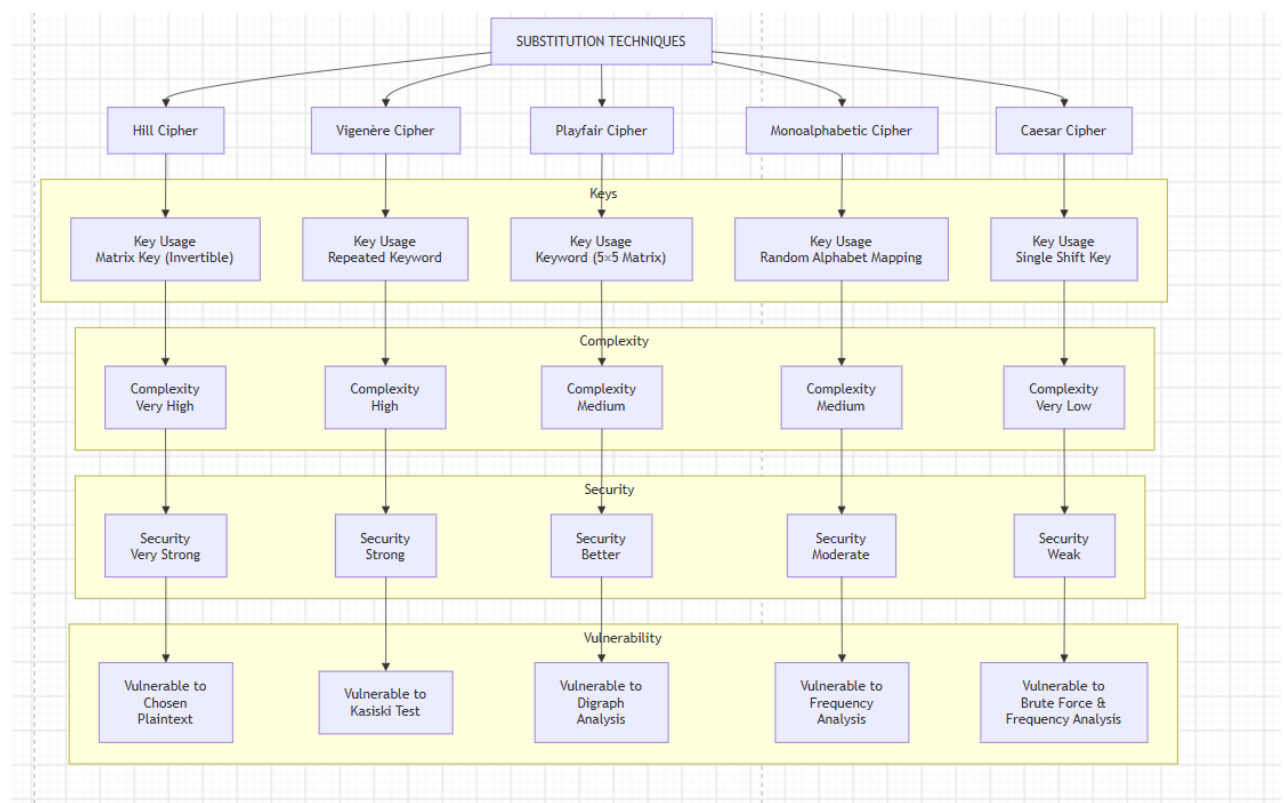
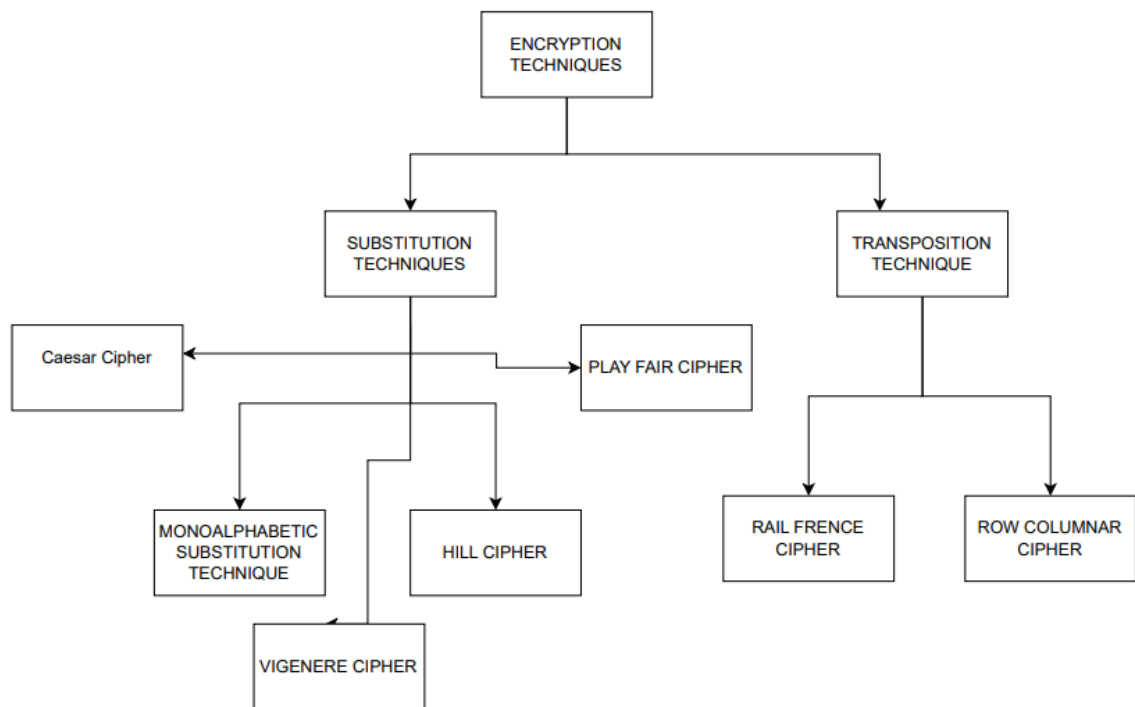
SLOUTIONS:

2) Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

ANSWER:

Concept Map – Comparison of Substitution Techniques

Concept map:



Explanation

Introduction

Substitution techniques are one of the oldest methods of encryption. In these techniques, each letter or group of letters in the plaintext is replaced with another letter, symbol, or group of letters to produce ciphertext. Different substitution techniques vary in terms of key usage, complexity, and security strength.

1. Caesar Cipher

Key Usage

- Uses a single numerical key (0–25).
- Every character is shifted by the same number of positions.

Complexity

- Very low.
- Easy to implement manually.

Security Strength

- Weak.
- Only 25 possible keys.

Vulnerability

- Easily broken using brute-force attacks.
- Frequency analysis also breaks it quickly.

Example

Plain Text : HELLO

Key : 3

Cipher Text : KHOOR

2. Monoalphabetic Cipher

Key Usage

- Uses one randomly arranged alphabet.
- Every letter is replaced by a fixed substitute.

Complexity

- Medium.
- Requires maintaining a substitution table.

Security Strength

- Moderate.
- Larger key space than Caesar Cipher.

Vulnerability

- Frequency analysis can still reveal the plaintext because letter frequencies remain unchanged.

Example

A → Q

B → L

C → P

3. Playfair Cipher

Key Usage

- Uses a keyword to generate a 5×5 matrix.
- Encrypts pairs of letters (digraphs).

Complexity

- Medium.
- More complex than Monoalphabetic Cipher.

Security Strength

- Better than Monoalphabetic Cipher because it encrypts two letters together.

Vulnerability

- Susceptible to digraph frequency analysis.

Example

Keyword : SECURITY

SECUR

ITYAB

DFGHK

LMNOP

QVWXZ

4. Vigenère Cipher

Key Usage

- Uses a repeated keyword.
- Each plaintext letter is encrypted using a different Caesar shift.

Complexity

- High.
- Multiple substitution alphabets are used.

Security Strength

- Strong.
- Frequency distribution is hidden better than in simple substitution.

Vulnerability

- Can be broken using Kasiski Examination or Index of Coincidence if the keyword repeats.

Example

Keyword : KEY

Plain Text : HELLO

Cipher Text : RIJVS

5. Hill Cipher

Key Usage

- Uses an invertible matrix as the key.
- Encrypts blocks of letters.

Complexity

- Very high.
- Requires matrix multiplication and modular arithmetic.

Security Strength

- Very strong among classical substitution ciphers.

Vulnerability

- Can be broken mainly through chosen-plaintext attacks if enough plaintext–ciphertext pairs are available.

Example

Key Matrix

|3 3|

|2 5|

Comparison Table

Technique	Key Usage	Complexity	Security Strength
Caesar Cipher	Single Shift Key	Very Low	Weak
Monoalphabetic Cipher	Random Alphabet	Medium	Moderate
Playfair Cipher	5×5 Matrix	Medium	Better
Vigenère Cipher	Repeated Keyword	High	Strong
Hill Cipher	Matrix Key	Very High	Very Strong

Security Order

Caesar Cipher
↓
Monoalphabetic Cipher
↓
Playfair Cipher
↓
Vigenère Cipher
↓
Hill Cipher

Analysis: Which Technique is Most Secure and Why?

Among the five substitution techniques, the **Hill Cipher** is considered the most secure. It encrypts blocks of letters instead of individual letters and uses matrix multiplication with modular arithmetic, making the relationship between plaintext and ciphertext highly complex. Unlike Caesar, Monoalphabetic, and Playfair ciphers, the Hill Cipher provides high diffusion, meaning that changing one letter in the plaintext affects multiple letters in the ciphertext. It is also resistant to simple frequency analysis because encryption is performed on groups of characters rather than individual characters.

The **Vigenère Cipher** is more secure than Caesar, Monoalphabetic, and Playfair because it uses multiple substitution alphabets, making frequency analysis much more difficult. However, if the keyword repeats, attacks such as the **Kasiski Examination** can reveal the key length and eventually break the cipher.

The **Playfair Cipher** improves security by encrypting pairs of letters instead of single letters, reducing the effectiveness of single-letter frequency analysis. Nevertheless, it is still vulnerable to digraph frequency analysis.

The **Monoalphabetic Cipher** has a much larger key space than the Caesar Cipher, but because each plaintext letter always maps to the same ciphertext letter, the frequency distribution of letters remains unchanged. As a result, frequency analysis can still successfully break the cipher.

The **Caesar Cipher** provides the lowest level of security because it uses only one shift value, giving an attacker only 25 possible keys to test. It can be broken almost instantly using brute-force attacks or frequency analysis.

Therefore, based on **key usage, complexity, resistance to attacks, and overall security**, the order from weakest to strongest is:

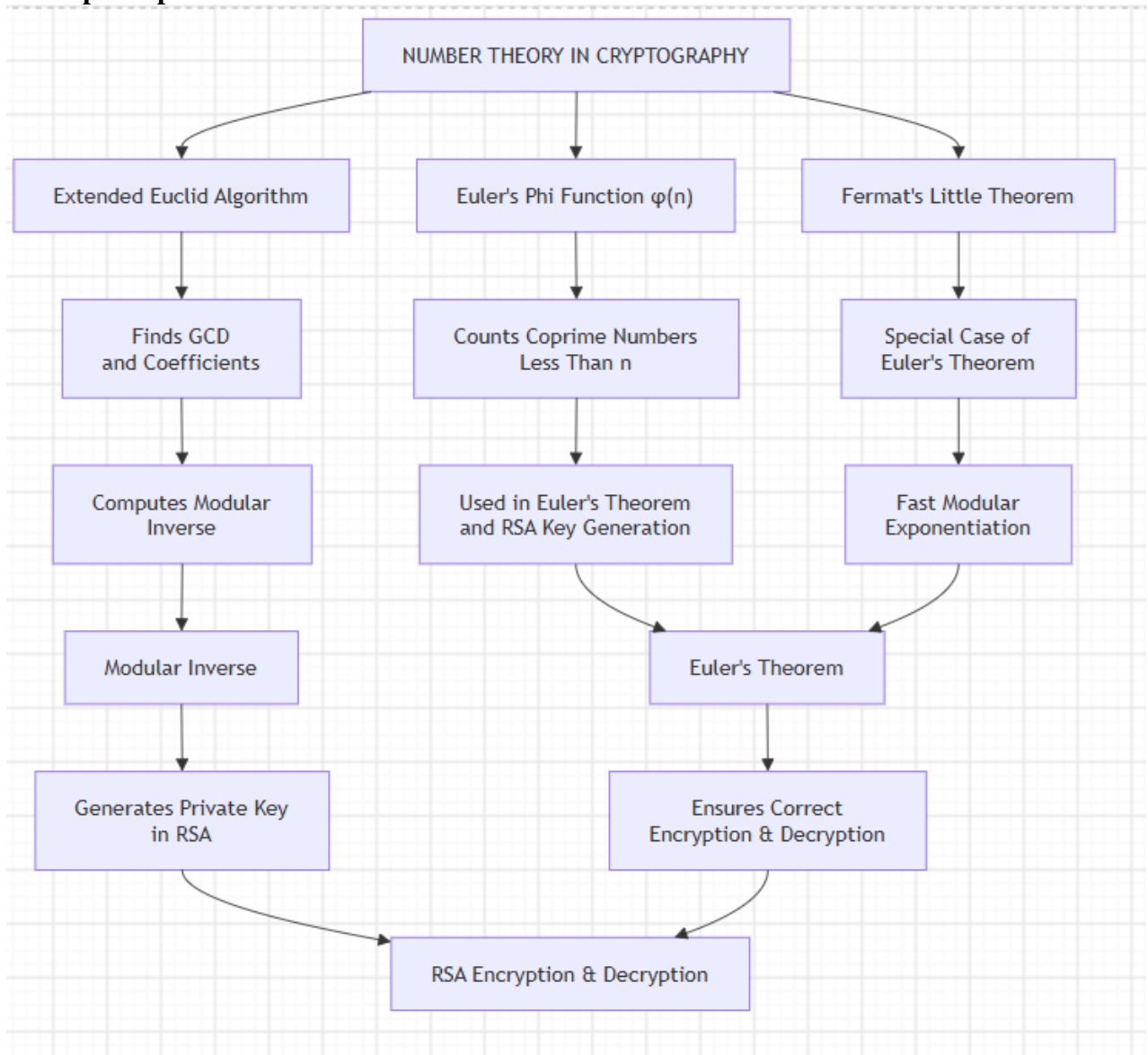
Caesar Cipher → Monoalphabetic Cipher → Playfair Cipher → Vigenère Cipher → Hill Cipher

The **Hill Cipher** is the strongest among the classical substitution techniques because of its mathematical foundation, block encryption method, and greater resistance to traditional cryptanalytic attacks.

5) Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:

- Show how each concept is mathematically connected
- Map their role in encryption/decryption
- Analyze which concept is most critical for public key cryptography and justify

Concept Map – Number Theory Concepts Used in Cryptography
Concept Map



Introduction

Modern cryptography depends on mathematical concepts from number theory. These concepts provide the foundation for secure encryption, decryption, digital signatures, and key generation. Public-key cryptographic algorithms such as RSA rely heavily on the Extended Euclid Algorithm, Modular Inverse, Euler's Phi Function, Euler's Theorem, and Fermat's Little Theorem. These concepts work together to ensure secure communication over insecure networks.

1. Extended Euclid Algorithm

Definition

The Extended Euclid Algorithm is an extension of Euclid's Algorithm. It not only finds the Greatest Common Divisor (GCD) of two integers but also computes integers x and y that satisfy:

$$ax + by = \gcd(a, b)$$

Mathematical Connection

- Finds the GCD of two numbers.
- Determines whether a modular inverse exists.

- Computes the modular inverse when $\gcd(a, n) = 1$.

Role in Encryption and Decryption

- Used during RSA key generation.
- Calculates the private key.
- Helps compute the modular inverse of the public exponent.

Example

If

$$\gcd(17, 3120) = 1$$

then the Extended Euclid Algorithm finds the modular inverse of 17 modulo 3120, which becomes the RSA private key.

2. Modular Inverse

Definition

The modular inverse of **a** modulo **n** is a number **b** such that

$$a \times b \equiv 1 \pmod{n}$$

It exists only when

$$\gcd(a, n) = 1$$

Mathematical Connection

- Depends on the Extended Euclid Algorithm.
- Used after calculating Euler's Phi Function.
- Essential for RSA private key generation.

Role in Encryption and Decryption

- Generates the private exponent (d).
- Allows encrypted data to be decrypted.
- Used in RSA and Digital Signature Algorithms.

Example

$$7 \times 15 = 105$$

$$105 \bmod 26 = 1$$

Therefore,

15 is the modular inverse of 7 modulo 26.

3. Euler's Phi Function

Definition

Euler's Phi Function, denoted by $\phi(n)$, counts the number of positive integers less than **n** that are relatively prime to **n**.

Mathematical Connection

For prime numbers,

$$\phi(p) = p - 1$$

For RSA,

$$\phi(n) = (p-1)(q-1)$$

where **p** and **q** are prime numbers.

Role in Encryption and Decryption

- Computes RSA parameters.
- Determines the size of the multiplicative group.
- Used to calculate public and private keys.

Example

$$\phi(9) = 6$$

The numbers

$$1, 2, 4, 5, 7, 8$$

are relatively prime to 9.

4. Euler's Theorem

Definition

Euler's Theorem states that

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

when

$$\gcd(a, n) = 1$$

Mathematical Connection

- Uses Euler's Phi Function.
- Generalizes Fermat's Little Theorem.
- Fundamental theorem behind RSA.

Role in Encryption and Decryption

- Ensures correct decryption.
- Guarantees that encrypted messages can be recovered using the private key.

Example

$$a = 3$$

$$n = 10$$

$$\phi(10)=4$$

Therefore,

$$3^4 = 81$$

$$81 \bmod 10 = 1$$

Hence Euler's theorem is satisfied.

5. Fermat's Little Theorem

Definition

If p is prime,

$$a^{(p-1)} \equiv 1 \pmod{p}$$

provided

$$\gcd(a, p) = 1$$

Mathematical Connection

- Special case of Euler's Theorem.
- Used for prime numbers.
- Helps simplify modular arithmetic.

Role in Encryption and Decryption

- Used in modular exponentiation.
- Helps perform efficient calculations.
- Used in primality testing.

Example

Take

$$a=2$$

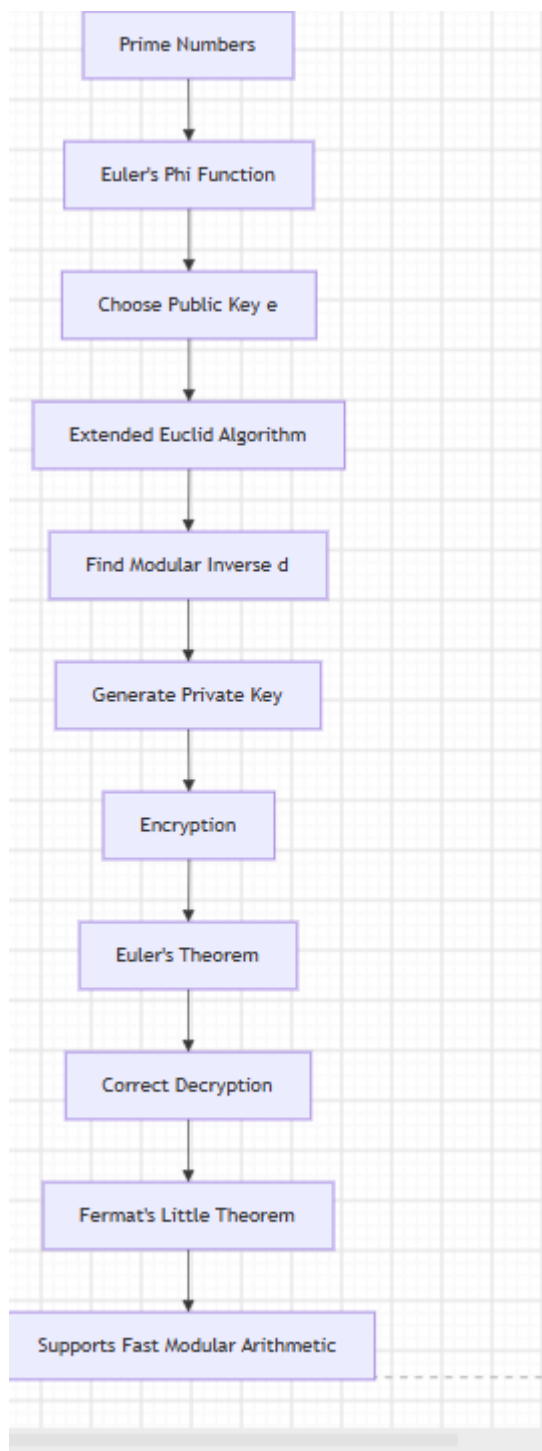
$$p=7$$

$$2^6=64$$

$$64 \bmod 7 = 1$$

Hence Fermat's Little Theorem is verified.

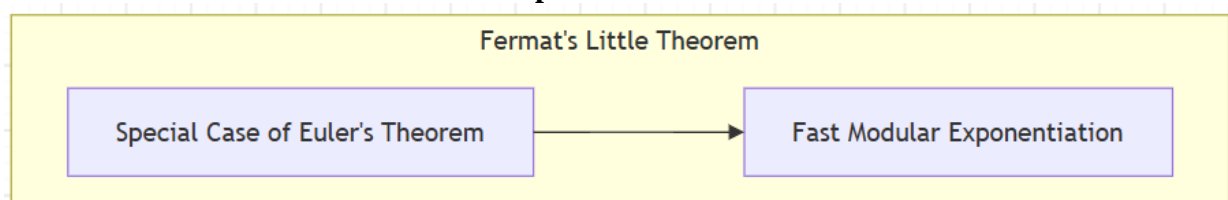
Relationship Between the Concepts:

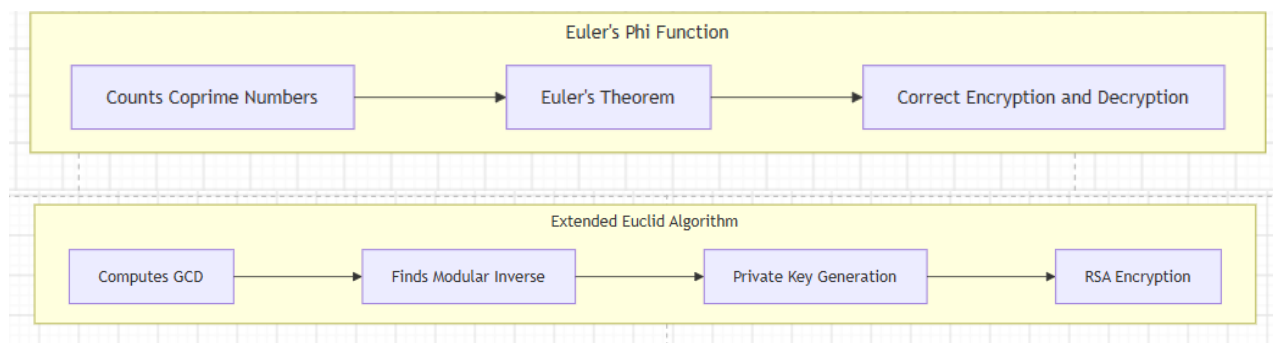


Role of Each Concept in Cryptography

Concept	Mathematical Purpose	Role in Cryptography
Extended Euclid Algorithm	Finds GCD and coefficients	Computes modular inverse for RSA
Modular Inverse	Finds inverse modulo n	Generates RSA private key
Euler's Phi Function	Counts coprime integers	Computes RSA parameters
Euler's Theorem	Modular arithmetic theorem	Ensures successful decryption
Fermat's Little Theorem	Special case of Euler's theorem	Fast modular arithmetic and primality testing

Mathematical Connection of All Concepts





Analysis: Which Concept is Most Critical for Public-Key Cryptography?

Among all the number theory concepts, Modular Inverse is the most critical for public-key cryptography because it is directly responsible for generating the private key in RSA. During RSA key generation, the public exponent (e) is selected first, and the private exponent (d) is calculated as the modular inverse of e with respect to Euler's Phi Function. Without the modular inverse, the private key cannot be generated, making decryption and digital signatures impossible.

The Extended Euclid Algorithm is important because it computes the modular inverse efficiently, but its primary role is to support the calculation of the inverse. Euler's Phi Function determines the number of integers that are relatively prime to the modulus and is used to derive RSA parameters. Euler's Theorem provides the mathematical proof that decryption correctly recovers the original message after encryption. Fermat's Little Theorem is a simplified version of Euler's Theorem that works for prime numbers and is useful for modular arithmetic and primality testing.

Therefore, Modular Inverse is considered the cornerstone of RSA because it directly enables the creation of the private key required for secure decryption and digital signatures. While the other concepts provide mathematical support, the modular inverse is the key element that makes public-key cryptography practical and secure.

Conclusion

Number theory forms the mathematical backbone of modern cryptography. The Extended Euclid Algorithm, Modular Inverse, Euler's Phi Function, Euler's Theorem, and Fermat's Little Theorem work together to enable secure key generation, encryption, and decryption in algorithms such as RSA. Among these concepts, the Modular Inverse plays the most significant role because it generates the private key, making secure communication, authentication, and digital signatures possible. Together, these mathematical principles ensure confidentiality, integrity, and secure communication over digital networks.